

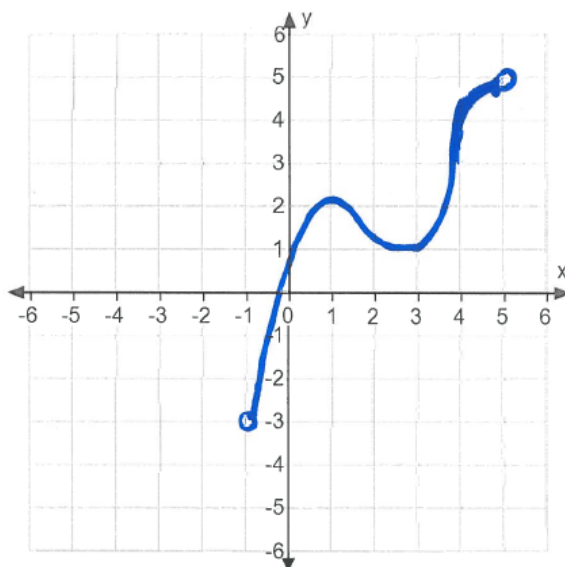
Problem 2

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Problem 1 In each problem, sketch a graph of a function meeting the given criteria.

- (a) Sketch a possible graph of a function which is continuous on an open interval $(-1, 5)$ but does not have a global maximum or minimum.

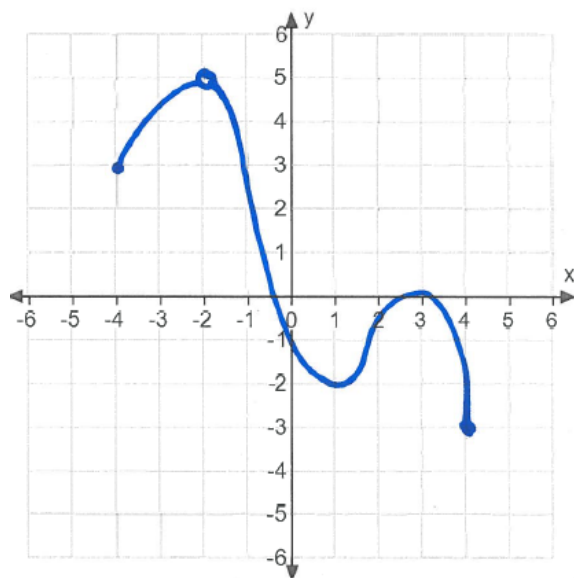
Explanation.



- (b) Sketch a possible graph of a function with a global minimum, a local maximum and a local minimum, but no global maximum on the interval $[-4, 5]$.

Explanation.

Problem 2



- (c) Sketch a possible graph of a function f , continuous on $[1, 4]$ with the following properties: $f'(x) = 0$ for $x = 2$ and $x = 3$; f has a global minimum at $x = 4$; f has a global maximum at $x = 3$, and f has a local minimum at $x = 2$.

Explanation.

